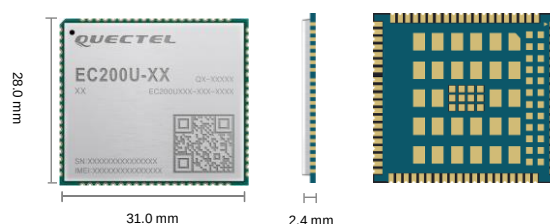


# Quectel EC200U Series

## IoT/M2M-optimized LTE Cat 1 Module



Quectel EC200U series is a series of the latest LTE Cat 1 module optimized specially for M2M and IoT applications. It delivers maximum data rates up to 10 Mbps downlink and 5 Mbps uplink. Designed in the compact and unified form factor, EC200U series is compatible with Quectel multi-mode LTE Standard EC25 series/EC21 series/EC20 R2.1/EC20-CN/EC200T series/EC200S series/EC25-G/EC21-G and UMTS/HSPA+ UC200T series, ensuring that it can easily migrate from 3G to 4G network. EC200U series also supports standard Mini PCIe packages to meet the needs of different industry applications.

EC200U series contains two variants: EC200U-CN and EC200U-EU, which meets the frequency band coverage of different countries and regions. Adopting laser engraving process, EC200U series features a more fashionable appearance, strong metallic texture, better heat dissipation capability and durable label information, which makes it ideally suited for automation requirement.

EC200U series has a rich set of Internet protocols, industry-standard interfaces and abundant functionalities (USB drivers for Windows 7/8/8.1/10, Linux, Android) extend the applicability of the module to a wide range of M2M and IoT applications such as POS/POC/ETC, shared equipment, data card, energy control and monitoring, security and protection, and industrial PDA.



## Key Features

- ✓ LTE Cat 1 module optimized for M2M and IoT applications
- ✓ Worldwide LTE, GSM/GPRS coverage
- ✓ Abundant functional interfaces
- ✓ Support analog audio
- ✓ Built-in GNSS (optional)
- ✓ Feature refinements: support Wi-Fi Scan (optional), Bluetooth (optional) and FOTA
- ✓ High performance



LTE Cat 1  
Max. 10 Mbps (DL)  
Max. 5 Mbps (UL)



LCC Package



Embedded Abundant  
Protocols



FOTA



Bluetooth  
(Optional)



GNSS (Optional)



USB 2.0 High Speed  
Interface



Wi-Fi Scan  
(Optional)



Quectel Enhanced AT  
Commands

# Quectel EC200U Series

| LTE Cat 1                       | EC200U-CN                                         | EC200U-EU                                  |
|---------------------------------|---------------------------------------------------|--------------------------------------------|
| Region/Operator                 | China                                             | EMEA                                       |
| Dimensions (mm)                 | 28.0 × 31.0 × 2.4                                 | 28.0 × 31.0 × 2.4                          |
| Temperature Range               |                                                   |                                            |
| Operating Temperature           | -35 °C to +75 °C                                  | -35 °C to +75 °C                           |
| Extended Temperature            | -40 °C to +85 °C                                  | -40 °C to +85 °C                           |
| Frequency Bands                 |                                                   |                                            |
| LTE-FDD                         | B1/B3/B5/B8                                       | B1/B3/B5/B7/B8/B20/B28                     |
| LTE-TDD                         | B34/B38/B39/B40/B41                               | B38/B40/B41                                |
| GSM                             | B3/B8                                             | B2/B3/B5/B8                                |
| GNSS                            | GPS/BeiDou                                        | GPS/BeiDou/Galileo/GLONASS                 |
| BLE                             | Bluetooth 4.2 (BR/EDR)                            | Bluetooth 4.2 (BR/EDR)                     |
| Wi-Fi                           | 2.4 GHz 11b (RX)                                  | 2.4 GHz 11b (RX)                           |
| Bandwidth                       | 1.4/3/5/10/15/20 MHz                              | 1.4/3/5/10/15/20 MHz                       |
| Certifications                  |                                                   |                                            |
| Regulatory                      | China: SRRC/ NAL/ CCC                             | Europe: CE*<br>Australia/New Zealand: RCM* |
| Carrier                         | China: China Mobile/ China Unicom/ China Telecom* |                                            |
| Max. Data Rate                  |                                                   |                                            |
| LTE-FDD (Mbps)                  | 10 (DL)/ 5 (UL)                                   | 10 (DL)/ 5 (UL)                            |
| LTE-TDD (Mbps)                  | 8.96 (DL)/ 3.1 (UL)                               | 8.96 (DL)/ 3.1 (UL)                        |
| GSM (kbps)                      | 85.6 (DL)/ 85.6 (UL)                              | 85.6 (DL)/ 85.6 (UL)                       |
| Interfaces                      |                                                   |                                            |
| (U)SIM                          | × 1                                               | × 1                                        |
| UART                            | × 3                                               | × 3                                        |
| USB 2.0                         | × 1                                               | × 1                                        |
| Audio Digital (PCM)             | × 1                                               | × 1                                        |
| Audio Analog (Input)            | × 1                                               | × 1                                        |
| Audio Analog (Output)           | × 1                                               | × 1                                        |
| ADC                             | × 3                                               | × 3                                        |
| SPI                             | × 1                                               | × 1                                        |
| SD Card                         | × 1                                               | × 1                                        |
| RESET                           | × 1                                               | × 1                                        |
| PWERKEY                         | × 1                                               | × 1                                        |
| I2C                             | × 2                                               | × 2                                        |
| LCD                             | × 1                                               | × 1                                        |
| Camera                          | × 1 (QuecOpen®)                                   | × 1 (QuecOpen®)                            |
| Matrix Keyboard                 | × 1 (4 × 4)                                       | × 1 (4 × 4)                                |
| Wi-Fi Scan/Bluetooth Antenna    | × 1                                               | × 1                                        |
| GNSS Antenna                    | × 1                                               | × 1                                        |
| Main Antenna                    | × 1                                               | × 1                                        |
| Enhanced Features               |                                                   |                                            |
| FOTA                            | ●                                                 | ●                                          |
| BLE (Bluetooth 4.2)             | ○                                                 | ○                                          |
| Wi-Fi Scan                      | ○                                                 | ○                                          |
| GNSS                            | ○                                                 | ○                                          |
| Audio Playback/ Audio Recording | ●                                                 | ●                                          |
| QuecLocator®                    | ●                                                 | ●                                          |
| QuecPython®                     | ●                                                 | ●                                          |
| QuecFile®                       | ●                                                 | ●                                          |
| (U)SIM Card Detection           | ●                                                 | ●                                          |

Notes:  
1. \* : means under development/planning.  
2. ● : Supported.  
3. ○ : Optional.

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| LTE Cat 1                   | EC200U-CN                                                            | EC200U-EU                                                            |
|-----------------------------|----------------------------------------------------------------------|----------------------------------------------------------------------|
| Software Features           |                                                                      |                                                                      |
| Protocols                   | TCP/UDP/PPP/NITZ/PING/FILE/MQTT/NTP/HTTP/HTTPS/SSL/FTP/FTPS/CMUX/MMS | TCP/UDP/PPP/NITZ/PING/FILE/MQTT/NTP/HTTP/HTTPS/SSL/FTP/FTPS/CMUX/MMS |
| Driver                      | RIL Driver                                                           | Android 4.x–11.x                                                     |
|                             | USB RNDIS Driver                                                     | Windows 7/8/8.1/10,<br>Linux 2.6–5.12                                |
|                             | GNSS Driver                                                          | Android 4.x–11.x                                                     |
|                             | USB ECM Driver                                                       | Linux 2.6–5.12                                                       |
|                             | USB Serial Driver                                                    | Windows 7/8/8.1/10,<br>Linux 2.6–5.12,<br>Android 4.x–11.x           |
| Electrical Features         |                                                                      |                                                                      |
| Supply Voltage Range        | 3.3–4.3 V, 3.8 V Typ.                                                | 3.3–4.3 V, 3.8 V Typ.                                                |
| Power Consumption (Typical) | 12 $\mu$ A @ Power off                                               | 12 $\mu$ A @ Power off                                               |
|                             | 1.7 mA @ LTE sleep (PF = 128)                                        | 1.7 mA @ LTE sleep (PF = 128)                                        |
|                             | 1.4 mA @ LTE sleep (PF = 256)                                        | 1.4 mA @ LTE sleep (PF = 256)                                        |
|                             | 28 mA @ Idle (PF = 64, USB Connection)                               | 28 mA @ Idle (PF = 64, USB Connection)                               |
|                             | 13 mA @ Idle (PF = 64, USB Disconnection)                            | 13 mA @ Idle (PF = 64, USB Disconnection)                            |

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